

Database Concepts

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Joins

- Combine data from two or more tables
- INNER JOIN. (All rows present in both tables) - default
- LEFT OUTER JOIN - (all rows present in left table, matching rows in right table)
- RIGHT OUTER JOIN - (all rows present in right table, matching rows in left table)
- FULL OUTER JOIN - Not applicable in all databases.

Join syntax

```
SELECT columns  
FROM table1  
LEFT JOIN table2  
ON condition;
```

```
SELECT s.student_name, b.book_name  
FROM student s  
JOIN book_issues bi ON s.student_id = bi.student_id  
JOIN books b ON bi.book_id = b.book_id;
```

Left Outer join

- Show all the students and the books they issued (even if they haven't issued any books)

Right Outer join

- Show all the books and the issued students (even if the books have not been issued).

Views

- A view into a selected set of columns from one table or multiple tables
- Can be used for queries (if saved earlier)
- Retrieves results by running the query
- Uses:
 1. Security - Hide sensitive columns from base tables.
 2. No need to rewrite queries with joins
 3. Data table independence

OR, AND, IN, WHERE

- OR - one of the conditions must be true - to expand the query results
- AND - all conditions must be true - narrow the query results
- IN - same as OR (simpler)
- WHERE - condition for queries, update, delete

UNION

- UNION - combines output of multiple queries (removes duplicates) as multiple rows
- UNION ALL - same as union, does not remove duplicates
- Same number of columns and datatypes to be returned in each query.
- Sytnax:

```
SELECT column_list FROM table1  
  
UNION  
  
SELECT column_list FROM table2;
```


ORDER BY, GROUP BY, distinct

- ORDER BY - Sort functionality - ASC (default), DESC
- GROUP BY - for aggregating rows corresponding to a column - apply functions such as COUNT, MAX, MIN, AVG
- Distinct - Unique records

Functions

- Max
- Min
- Avg
- Count
- Sum

Sub-Queries

- Multiple queries where one queries output is fed to another query.
- Display the name of the student with max fine

Auto_increment

- Auto assigns values
- Increments the previous value of the column by 1
- Only one auto_increment column per table
- Increment delta can be changed as a global setting : `SET GLOBAL auto_increment_increment = 5;`